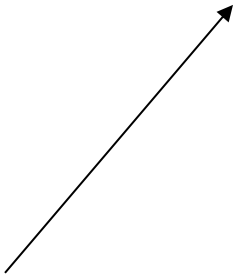
Compiled Problems – Senior Division

1. **11-12 C1 Assembly Language**

What is the final value of N when the program is run?

```
A     DC      32          LOAD    C
B     DC      2           STORE   A
N     DC      0           BG      TOP
TOP   LOAD    A           LOAD    N
      DIV     B           SUB     =1
      STORE   C           STORE   N
      LOAD    N           PRINT   N
      ADD     =1          END
      STORE   N
```



2. **12-13 C1 Assembly Language**

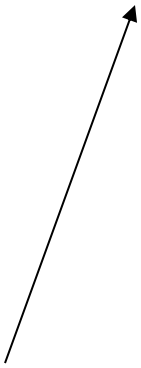
What is the final value of S that is printed when the program is run?

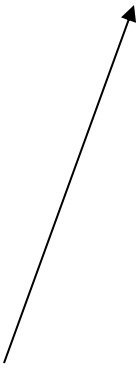
```
S     DC      0
N     DC      10
TOP   LOAD    N
      MULT    N
      ADD     S
      STORE   S
      LOAD    N
      SUB     =1
      STORE   N
      BG      TOP
      PRINT   S
      END
```

3. **13-14 C1 Assembly Language**

What is the final value of E when the program is run?

```
A     DC      10          T     LOAD    A           P     LOAD    A
B     DC      2                 SUB     D                 SUB     B
C     DC      5                 BE      U                 STORE   A
D     DC      8           V     LOAD    A                 BU      R
      LOAD    A                 DIV     B           S     LOAD    B
      SUB     B                 ADD     C                 ADD     C
      BG      P                 MULT    D                 STORE   B
R     LOAD    B                 STORE   E                 BU      T
      MULT    C                 LOAD    E           U     LOAD    D
      SUB     D                 PRINT   E                 SUB     D
      BL      S                 END                       STORE   D
                                                          BU      V
```

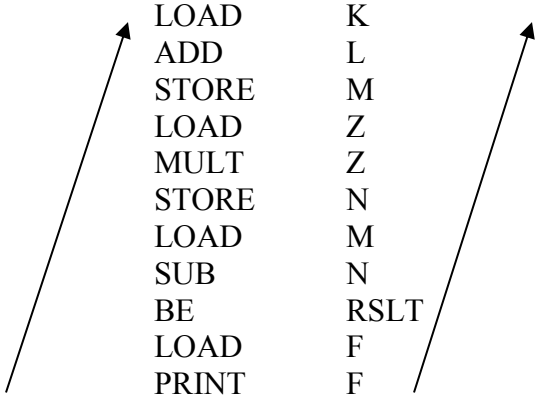




4. **14-15 C1 Assembly Language**

What is printed when the program is run?

```
X     DC      21        LOAD    K              BU      DONE
Y     DC      20        ADD     L       RSLT   LOAD    T
Z     DC      23        STORE   M              PRINT   T
T     DC      1         LOAD    Z       DONE   END
F     DC      0         MULT    Z
      LOAD    X         STORE   N
      MULT    X         LOAD    M
      STORE   K         SUB     N
      LOAD    Y         BE      RSLT
      MULT    Y         LOAD    F
      STORE   L         PRINT   F
```



5. 15-16 C1 Assembly Language

What is printed when the program is run?

```
A     DC      2         LOAD    D
B     DC      1         ADD     = 1
C     DC      1         STORE   D
D     DC      1         LOAD    C
TOP   LOAD    B         SUB     = 100
      MULT    A         BL      TOP
      STORE   B         PRINT   D
      LOAD    C         END
      ADD     B
      STORE   C
```

